Macaque Re-identification — All-policy Comparison

28 October 2026 · revised 30 October 2026 · supervisor review

149 identities · 6,665 train / 1,302 validation / 1,498 test crops · one fit per policy

1. Findings for supervisor discussion

±45° training with seven-angle mean probability averaging is the current validation-based choice. It identifies **1,275/1,498 test images (85.11%)**, with **83.74% macro F1**. This is **0.60 percentage points / nine images** above ±30° mean and **2.60 points / 39 images** above the original baseline without TTA.

The comparison supports three distinct findings:

- **Single-image performance:** ±30° has the highest validation top-1 (82.49%). ±30° and ±45° tie on single-image test top-1 (83.51%); ±60° scores 83.11%.
- **Averaging:** ±45° mean reaches 85.11% test top-1; ±60° mean reaches 84.98%; ±30° mean reaches 84.51%. Both ±30° mean and ±45° mean score 83.56% on validation; ±45° has higher validation macro F1 (80.11% versus 79.82%).
- **Strength and reliability:** ±100° and ±150° have lower test top-1 at this 30-epoch budget. The ±30°/±45°/±60° differences are small and have not been confirmed across training seeds.

Training policy	Validation single	Validation mean	Test single	Test mean	Test mean macro F1
Original baseline	80.49%	Not evaluated	82.51%	Not evaluated	Not evaluated
±30°	82.49%	83.56%	83.51%	84.51%	82.84%
±45°	82.10%	83.56%	83.51%	85.11%	83.74%
±60°	82.18%	83.41%	83.11%	84.98%	83.37%
±100°	79.49%	81.26%	81.44%	82.44%	79.99%
±150°	56.30%	58.06%	56.94%	58.41%	46.36%

All percentages above refer to top-1 accuracy except the final F1 column. “Single” uses one unrotated image. “Mean” averages seven probability vectors at −30°, −20°, −10°, 0°, +10°, +20°, +30°. Baseline TTA was not evaluated. The original baseline uses affine ±15° plus possible RandAugment rotation.

One-page comparison sheet for the meeting (landscape PDF).

2. Training and validation learning curves — all policies together

The same policy colours and ordering are used throughout this report. Both loss panels show **all six policies, all 30 epochs, without smoothing**, on a common **0–20 loss scale**. This replaces the separate original/follow-up comparison in the main results presentation.

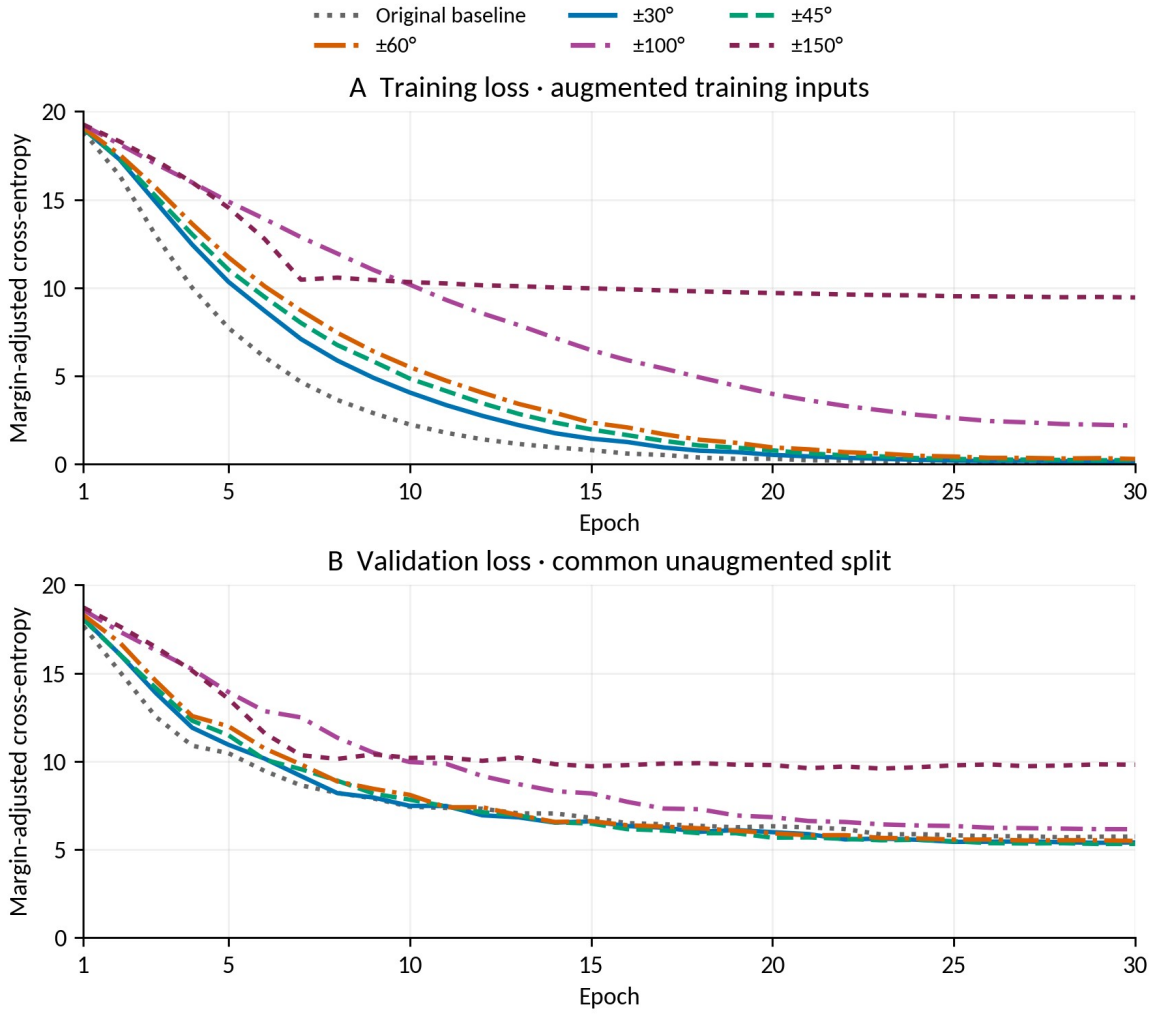


Figure 1. Shared learning-curve comparison. Panel A shows training loss on each policy's augmented inputs; panel B shows validation loss on the same clean validation cohort. Training inputs differ in difficulty, so training loss alone does not rank generalisation. Both logged losses use cross-entropy with the label-dependent ArcFace training margin. Checkpoints were selected by ordinary validation top-1, not minimum loss. Full-size PDF.

3. Final held-out test comparison — accuracy and cross-entropy

The test set provides one final comparison per saved checkpoint and inference method. **No epoch-by-epoch test-loss series was recorded.** The figure therefore shows final test accuracy and final test cross-entropy as checkpoint summaries. It includes all three inference methods for every evaluated policy.

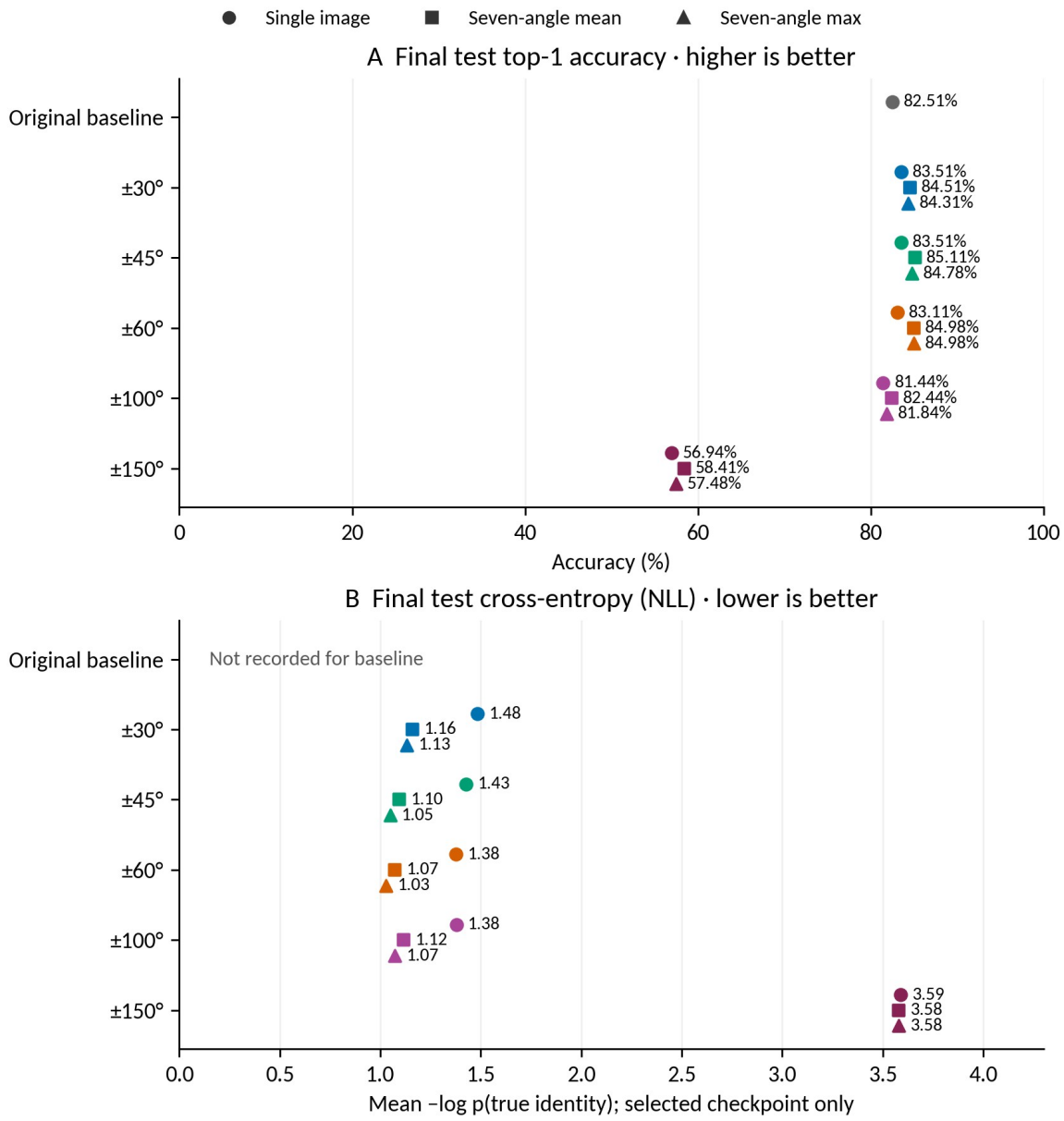


Figure 2. Final test comparison on the same 1,498 images. Higher top-1 is better; lower cross-entropy is better. Marker shape identifies inference method; policy colours match Figure 1. Baseline test loss is not recorded, and its TTA methods are not evaluated; neither missing result is represented as zero. Full-size PDF.

Detailed final test results

Training policy	Single top-1	Mean top-1	Max top-1	Mean macro F1
Original baseline	82.51%	Not evaluated	Not evaluated	Not evaluated
±30°	83.51%	84.51%	84.31%	82.84%
±45°	83.51%	85.11%	84.78%	83.74%
±60°	83.11%	84.98%	84.98%	83.37%
±100°	81.44%	82.44%	81.84%	79.99%
±150°	56.94%	58.41%	57.48%	46.36%

Test cross-entropy (NLL), lower is better

Training policy	Single NLL	Mean NLL	Max NLL
Original baseline	Not recorded	Not evaluated	Not evaluated
±30°	1.4832	1.1607	1.1313
±45°	1.4272	1.0951	1.0486
±60°	1.3775	1.0730	1.0283
±100°	1.3789	1.1162	1.0730
±150°	3.5889	3.5798	3.5799

Loss values and checkpoint timing

Training policy	Selected epoch	Train CE at epoch 30	Validation CE at epoch 30	Test NLL single	Test NLL mean
Original baseline	28	0.0959	5.7418	Not recorded	Not evaluated
±30°	29	0.1535	5.4072	1.4832	1.1607
±45°	27	0.2350	5.3246	1.4272	1.0951
±60°	30	0.2986	5.4917	1.3775	1.0730
±100°	30	2.1910	6.1596	1.3789	1.1162
±150°	30	9.4555	9.8135	3.5889	3.5798

The test columns describe the selected checkpoint; the learning-loss columns describe epoch 30. Train/validation CE includes the ArcFace margin; final test NLL uses the margin-free prediction probabilities. They are reported together for completeness but use separate plot panels and definitions. Their numerical magnitudes should not be read as a directly comparable train–test loss gap.

Test cross-entropy CSV · All source hashes, checks and curve values.

4. Augmentation settings and meeting examples

The previous `strong` profile explicitly used **RandomAffine(degrees=15)**, which samples an affine rotation from **−15° to +15°**. It then applied restricted RandAugment with two operations and magnitude 5. In the installed torchvision 0.17.2 implementation, a selected `Rotate` operation contributes **+5° or −5°**; operations are sampled sequentially and can repeat. The total baseline rotation could therefore exceed ±15°. It should not be described as an unaugmented or strictly ±15° baseline.

The new policies use one affine rotation step and do not add RandAugment rotations. The names ±30°, ±45°, ±60°, ±100° and ±150° describe training angle ranges, not fixed rotations. Only rotation differs among the ±30°/±45°/±60° policies; the ±100°/±150° policies also change other augmentation settings.

Policy	Training rotation	Affine scale	Translation	Noise	Gaussian blur	Random erasing
Baseline	Affine ±15°; RandAugment may add rotations	0.90–1.10	±8%	No explicit Gaussian noise	35%; σ 0.1–2.0; 5×5	30%; area 2–12%
±30°	Single affine: −30° to +30°	0.80–1.25	±10%	35%; σ 0.03	40%; σ 0.2–2.0; 7×7	20%; area 2–10%
±45°	Single affine: −45° to +45°	0.80–1.25	±10%	35%; σ 0.03	40%; σ 0.2–2.0; 7×7	20%; area 2–10%
±60°	Single affine: −60° to +60°	0.80–1.25	±10%	35%; σ 0.03	40%; σ 0.2–2.0; 7×7	20%; area 2–10%
±100°	Single affine: −100° to +100°	0.70–1.35	±12%	45%; σ 0.05	50%; σ 0.2–2.5; 7×7	25%; area 2–12%
±150°	Single affine: −150° to +150°	0.60–1.50	±15%	50%; σ 0.08	60%; σ 0.3–3.0; 7×7	30%; area 2–15%

The baseline also uses a random resized crop (area scale 0.8–1.0) and restricted RandAugment. The new policies resize to 224×224 before the affine operation and use explicit colour jitter instead of RandAugment. None of these six policies uses horizontal mirroring. Noise sigma is specified for float images in [0,1] before normalization. Scale and translation in the baseline row describe its explicit affine step; its other geometry operations can add further variation.

At test time, TTA uses exactly **−30°, −20°, −10°, 0°, +10°, +20°, +30°**, regardless of the training policy. Each view produces margin-free class logits followed by softmax. Mean pools the seven probability vectors; max takes each class's maximum view probability and renormalizes. TTA does not retrain or average model weights.

Implementation references: local transforms, RandomAffine 0.17 documentation, and RandAugment 0.17.2 source.

Meeting examples: original and augmented training images

For the meeting discussion, the following sheets show **real training-set crops alongside examples generated with the actual training augmentation code**. They use Abby, Harry and Warty; the sources are training images, not validation/test images chosen for their prediction results.

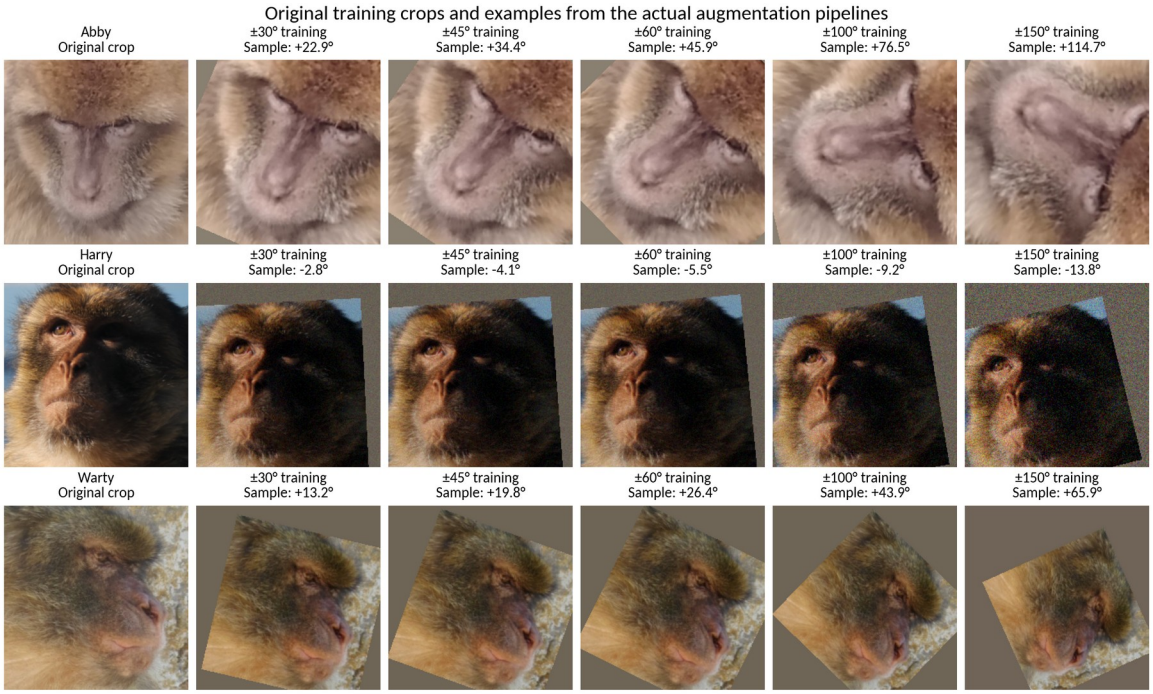


Figure 3. Original training crops and sampled full-pipeline augmentation. Each row uses the same source crop across the five policies.

How to read the training sheet. Each row starts with the same original face crop, resized to 224×224 without augmentation for display. The other columns are random samples from the full training pipelines, including affine rotation/zoom/ translation, colour jitter and probabilistic blur, noise and erasing. Column headings give the configured rotation range; the sampled rotation is printed below it. The ±30°/±45°/±60° policies share all other settings, while ±100°/±150° also increase other augmentation strengths. The same seed is reset across policies within each row (42, 43 and 44), so the illustrations can be reproduced.

These are **newly generated illustrations**, not exact images recovered from past training batches. Normalization is reversed and values are clipped to [0,1] for viewing. Grey affine borders and any erased patches are pipeline outputs. No horizontal mirroring is used. The source originals are existing detection crops rather than full camera photographs.

Training sheet: full-size PNG · PDF for presenting or printing · Individual originals and augmented images

Meeting examples: the seven inference views used for averaging

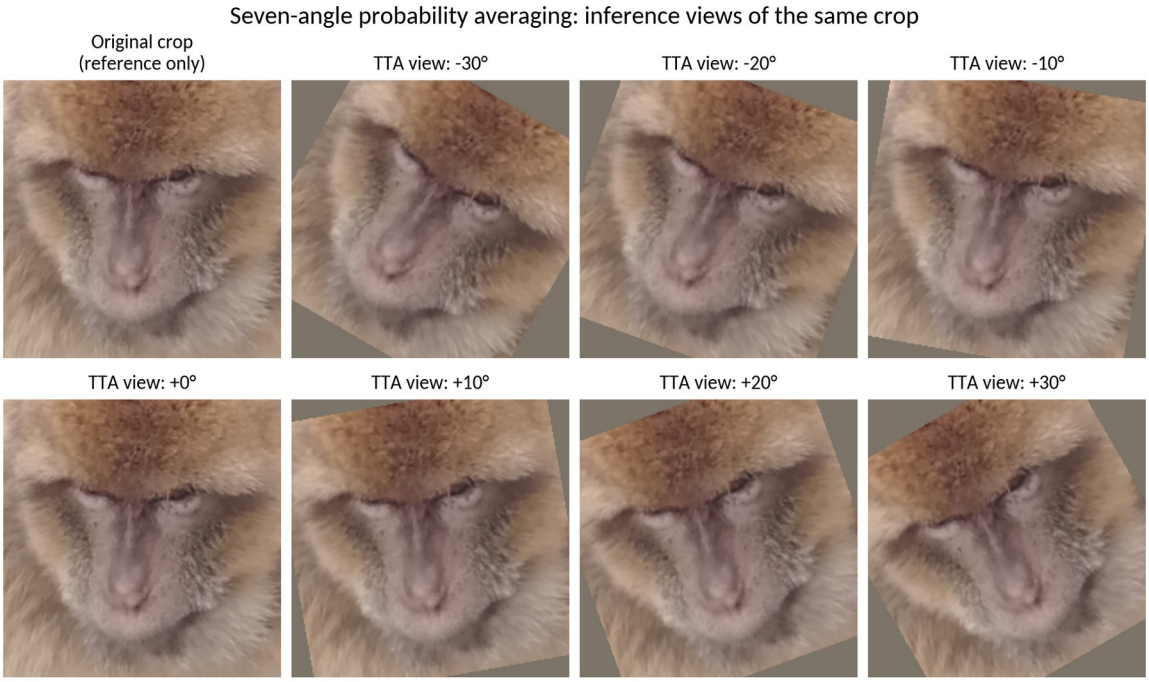


Figure 4. Seven deterministic inference views and an original reference. The reference is not an eighth inference view.

This second sheet illustrates **test-time augmentation (TTA)** using the Abby training crop only as a demonstration image. The original panel is a reference; it is not an eighth inference view. The seven views include 0° exactly once. Each view is resized/rotated and scored by the same saved model; their **class probabilities are averaged**, not their pixels or model weights. TTA does not apply the random training colour jitter, noise, blur or erasing. Its seven angles remain ±30° through 0° even for models trained with ±45° or ±60°.

Inference-view sheet: full-size PNG · PDF for presenting or printing · Source paths, image hashes, seeds and transform parameters · Reproduction script

5. Interpretation and individual-level results

Moderate augmentation and seven-angle averaging perform best in this experiment. The ±45° mean model correctly identifies 1,275 test images, compared with 1,266 for ±30° mean and 1,236 for the original baseline. Its averaging gain over its own single-image result is **1.60 points / 24 images**. Image-by-image comparison with the baseline fixes **72 errors** and introduces **33 errors**, a net gain of 39. Macro precision is **85.15%** and macro recall is **84.09%**.

Metric trade-off. ±45° mean has the highest test top-1. ±60° mean has lower test NLL than ±45° mean (1.0730 versus 1.0951), and ±60° max has the lowest measured NLL (1.0283). NLL scores the probability assigned to the true identity, including confident mistakes; top-1 counts correct classifications. These test descriptions do not change the validation-based choice.

The ±150° policy finishes with only 59.29% accuracy on its augmented training inputs and 56.30% validation accuracy. The shared curves show that it has difficulty fitting this training task at the chosen budget. Longer training has not been tested. Strong rotation/zoom may crop away identity cues, and noise, blur and erasing may obscure them; this mechanism is an interpretation rather than a separately measured cause.

The ±30°/±45°/±60° follow-up changes only rotation range while holding other settings fixed. Comparisons with the baseline or ±100°/±150° change several augmentation settings together. Neither comparison establishes a robust optimal angle from a single fit per policy.

Per-identity results for ±45° + mean

- Per-class metrics: selected ±45° + mean per-class CSV
- Per-image predictions: selected model predictions and hard examples
- Full evaluation summary: selected model metrics JSON
- Confusion matrix: 149×149 matrix in NumPy format

The five lowest-recall identities for the selected configuration are:

Identity	Test images	Recall / per-class accuracy	F1
Caro	3	0.00%	0.00%
Thief	3	33.33%	50.00%
Harry	7	42.86%	54.55%
Racoon	4	50.00%	66.67%
Nathalie	9	55.56%	62.50%

One identity has zero test recall. Low-support identities remain sensitive to individual errors; overall accuracy alone does not describe all individuals equally. New embedding projections were not computed by this comparison job, so the baseline's t-SNE plot is not reused for the new model.

6. Limitations, selection and reproducibility

- **Selection:** saved checkpoints use the best ordinary validation top-1. The original job selected ±30° mean among nine model/method combinations; the follow-up selected ±45° mean among six. Both choices were locked before their test scoring. Across suites, the two mean methods tie at exactly 1,088/1,302 validation images; validation macro F1 favours ±45° under the recorded tie-break rule.
- **One fit per policy:** the measured nine-image advantage over ±30° mean is not a repeated-seed confidence interval or evidence of statistical significance. Test NLL is descriptive and was not used to change the selection.
- **Metric definitions:** training/validation loss includes the ArcFace training margin; final test NLL uses margin-free probabilities. No per-epoch test loss exists in these runs. Baseline test NLL is missing and baseline TTA was not evaluated.
- **Matched cohort:** all 15 augmented-model test prediction files use the baseline's exact 1,498 image/identity pairs. Both suites' split-manifest and registry SHA-256 pairs match. Registry date 14 October is not an experiment date.
- **Training control:** ImageNet-pretrained ResNet50, 256-dimensional embedding, ArcFace margin 0.5 / scale 30, AdamW LR 3e-4 / weight decay 0.01, cosine schedule, batch size 32, seed 42, GPU/AMP and 30 epochs. Logit adjustment was disabled (tau=0). Training accuracy is measured on augmented inputs of different difficulty.
- **Next evidence:** repeat the strongest candidates across seeds; evaluate baseline TTA if a matched inference comparison is needed. This documentation revision derives metrics from saved outputs and launches no additional model jobs.

The selected follow-up checkpoint is `artifacts/augmentation_sweeps/20261028T164412Z_JmVXbV/checkpoints/macaque-resnet50-arcface_aggressive_rot45_canonical_20261014_best.pt`.

Final test-loss calculation

How test loss was obtained. The original evaluation summaries report accuracy and F1, but do not store loss. For the five augmented models, this revision derives **NLL = mean[−ln p(true identity)]** from the saved test probabilities. Mean/max use the saved pooled probabilities; single-image NLL uses the cached 0° view. Every recovered prediction and top-1 confidence was checked against the corresponding recorded single/mean/max output. All true-class probabilities are positive; the calculation uses float64 logs without clipping. No model inference, training, new checkpoint selection or test-driven tuning was run.

The baseline's full probability arrays were not saved, so its NLL cannot be recovered from the available summaries. These final test NLL values are **margin-free**; the training/validation curves retain the **margin-adjusted** objective actually logged during fitting.

Appendix A. All-policy diagnostic loss panels

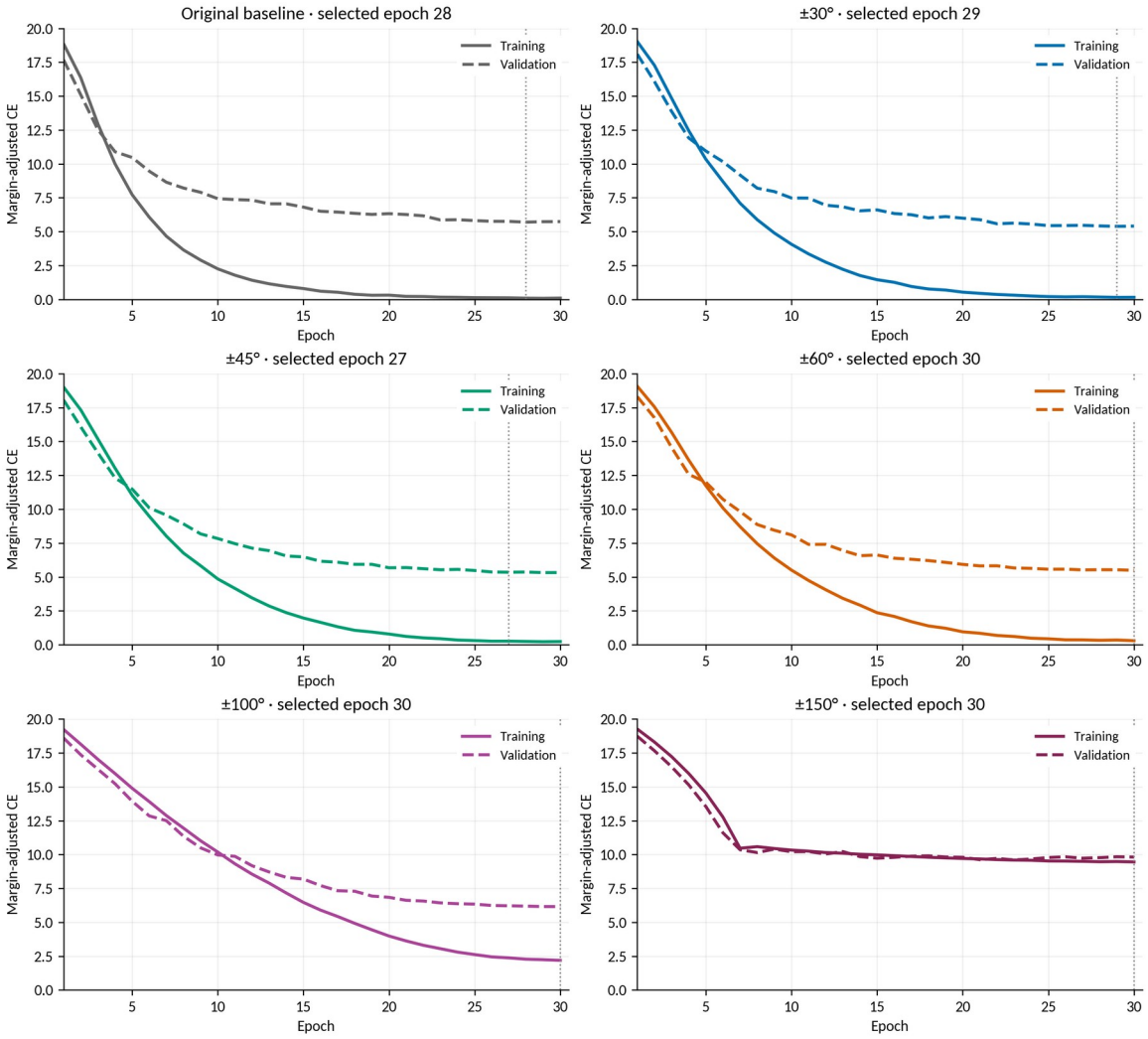


Figure 5. Per-policy detail using a common 0–20 loss scale. Solid lines show training, dashed lines validation, and vertical dotted lines the recorded selected epochs. The colours and policy ordering match the main comparison. Full-size diagnostic PDF.

Appendix B. All-policy training and validation accuracy

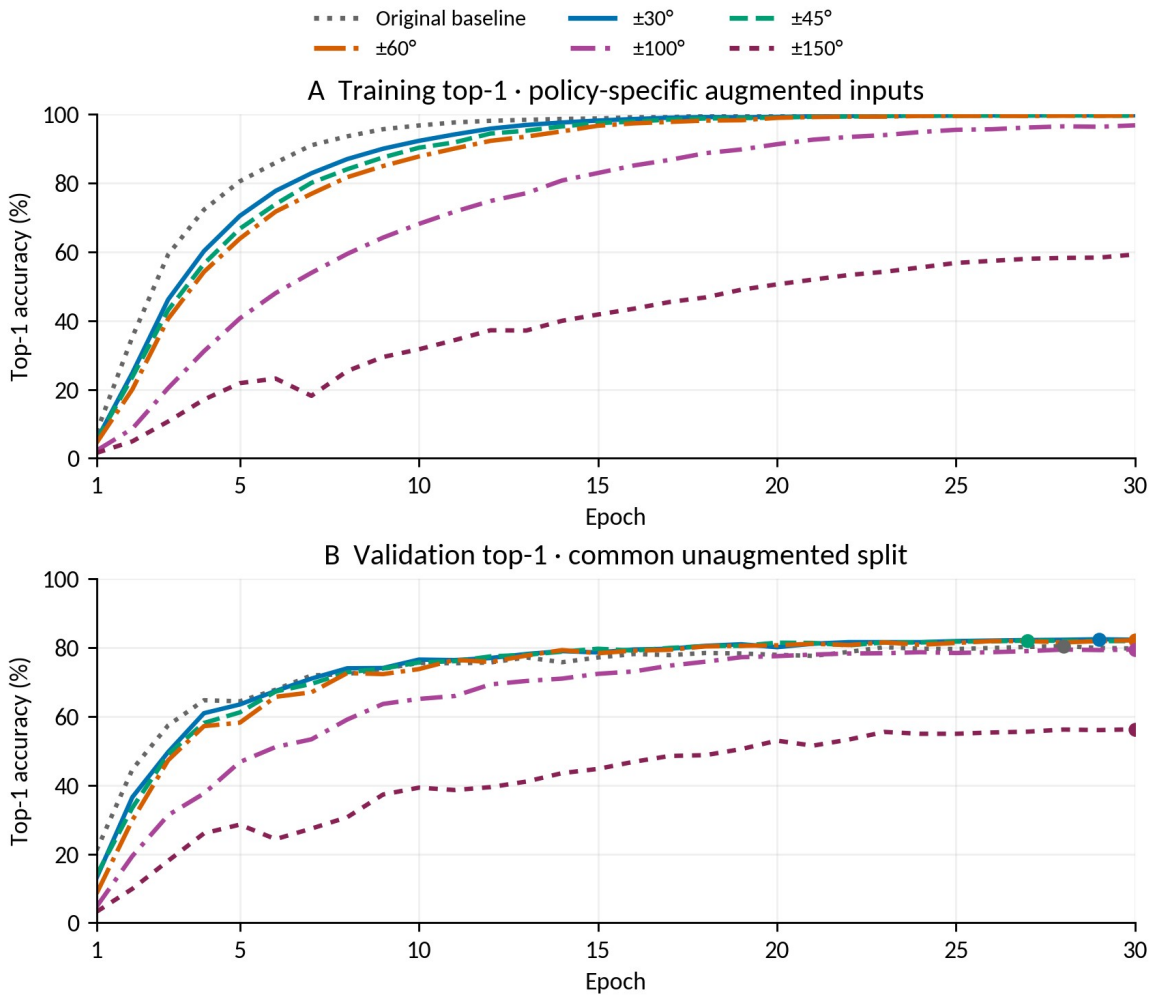


Figure 6. Corresponding accuracy curves for all policies. Dots mark the recorded selected epochs on the validation curves. Training accuracy uses policy-specific augmented inputs; validation uses the common clean split. Full-size accuracy PDF.

Appendix C. Source records and completed jobs

Source artifacts

- Combined curve values, source hashes and cached-probability checks.
- Reproduce the consolidated figures and final test NLL.
- Original 18 validation/test comparisons and follow-up 12 comparisons.
- Original selection record and follow-up selection record.
- Original data provenance and follow-up data provenance.
- Baseline evaluation report.

Completed jobs

Job	Task	Completion	Elapsed
38494871	Baseline training	23 October	24m 36s
39152252_0	±30° training	26 October	26m 12s
39152252_1	±100° training	26 October	12m 15s
39152252_2	±150° training	26 October	27m 54s
39152253	Original averaging comparison	28 October	17m 55s
39484293_0	±45° training	28 October	26m 24s
39484293_1	±60° training	28 October	26m 24s
39484294	Follow-up averaging comparison	29 October	30m 12s

All listed jobs completed with exit code 0:0. Wall times include startup and I/O and are not controlled compute-speed comparisons. Training/evaluation dates are in 2026. Source logs: original suite and follow-up suite.